Identification of adulterated organic fertilizer

Organic fertilizers are produced through decomposition of organic materials of plant and animal origin. The government of Bangladesh has approved the standard specification of organic fertilizer with certain physical and chemical properties through a Gazette Notification by the Ministry of Agriculture on 02 April 2008. As per notification the organic fertilizer should be-

- non-granular in form
- dark grey to black in colour
- odorless
- devoid of bad smell

A good quality organic fertilizer does not form clod when press in hand. These are the qualitative/physical properties of a standard organic fertilizer. By checking these properties preliminary idea about the quality of an organic fertilizer can be obtained. But it is not possible to be sure about the quality without chemical analysis in the laboratory.

5. Climate Smart Soil Management Practices

5.1 Soil organic matter management

Soil organic matter comes from plant and animal remains. It influences the physical, chemical and biological properties of soil. It improves soil physical conditions viz. soil structure, water holding capacity, aeration, soil erosion etc. It's a storehouse of plant nutrients, chiefly N, P & S. It serves as a food and energy for beneficial organisms viz. N_2 fixing bacteria (e.g *Rhizobium, Azobtobacter*), earthworms. It is the 'life force of a soil'.

Organic matter status

A good soil should have at least 2.5% organic matter, but in Bangladesh most of the soils have less than 1.5%, and some soils have even less than 1% organic matter. As the time advances, organic matter content in soil declines. This is particularly true under high land and medium high land conditions. The long-term fertilizer trials indicate that in the rice-rice (anaerobic-anaerobic) cropping system, the soil organic matter has slightly increased (BRRI and BAU reports) and in the wheat-rice (an aerobic-aerobic) system the soil organic matter has rather decreased (BARI report). Hence, depletion of soil organic matter cannot be generalized across the country. Organic matter status of Bangladesh soils is categorized as very low (0.408 mha, 4.7% arable land), low (2.58 mha, 30.1%) and medium (5.08 mha, 59.2%) (Hasan et al., 2020).

Organic matter mineralization

Soil organic matter undergoes mineralization and releases substantial quantities of N, P, S and smaller amount of micronutrients. Application of organic residues returns mineral nutrients to the soil. The conversion of organic N, P and S to available forms occurs through the activity of microorganisms and is influenced by temperature, moisture, pH etc. The rate and extent of mineralization determine crop availability of nutrients from added organic materials. All the bio-forms use soil as their home or they live on organic matter and decompose it to simple products. These products are responsible for sustaining soil productivity and performing environmental regulatory function.

Sources of soil organic matter

Soil organic matter is continuously undergoing changes and needs to be replenished regularly to maintain soil productivity. The major sources of soil organic matter include animal manure, farmyard wastes, domestic wastes, industrial wastes, sewage sludge, green manure etc. A large variety of organic wastes are available in the country that can be used as potential source of manure to improve soil. These are domestic wastes (non-edible vegetables, food and fruit parts, after-meal wastes etc.), farmyard wastes (cattle dung and urine, feed/fodder refuse, harvested crop residues, poultry excreta etc.), agro-industrial wastes (sugarcane trash, oil cakes, bagasse, molasses, bone meal, blood meal, fish meal, rice husk, brans, saw dust etc.), farm wastes (crop residues, weeds, dead animals, water hyacinth etc.) and city wastes (solid wastes and sewage sludge).

Crop residues

Leftover parts of various crops after harvest are called crop residues. Substantial quantities of crop residues are produced in the country every year. But little or no care is taken for its use. In most cases, crop residues are burnt or removed away to clean the land causing huge loss of this potential resource. Plant roots, straw & stalks and vegetable tops are valuable as a source of organic matter and plant nutrients. Crop residues can be recycled either by composting or by way of mulch or by direct incorporation in the soil.

Animal manure

It includes the excreta (dung and urine) of the domestic animals. Stubbles used as bedding of animals also become part of the manure. In Bangladesh, cowdung is the most common animal manure, although a big portion of the cowdung produced in the country is used as fuel. Next to cowdung, poultry manure is a potential source of organic matter.

Fresh animal manure should not be applied to standing crops, because the heat and CO_2 generated during vigorous decomposition is harmful for the young roots. Substantial quantities of animal manure and their nutrient content are lost due to careless handling. Animal manure should be stored in pits under a shade. The cattle urine is rich in N and should be preserved with the dung. The manure in the pit should be kept moist in order to reduce the volatilization of N. Animal manure generally takes 2-3 months for its decomposition.

Compost

The organic fertilizer that is produced by decomposing different waste materials of plant and animal origin is called compost. Ingredients that are used to make compost include dead leaves, straw, weeds, water hyacinth, household wastes like non-edible food, fruit and vegetable parts, after-meal wastes, municipal garbage, saw dust, wastes of leather factory, sugar mill bagasse, rice husk etc. Municipal and leather wastes should be treated to make them free from heavy metals and other toxic substances. The materials should be placed in layers, one above another. Each layer may be 25-30 cm thick. Heaps should preferably be 1.5 - 2.0 meter wide and not more than 1.5 meter in height. In order to promote microbial activities, thin (4-5 cm) layers of soil or fresh cowdung should be placed in between the layers of materials in the heap. Top of the heap should also be covered with soil. The heap should be kept moist, by spraying water at regular intervals. After 1.5 - 2 months the layers should be reversed in a new heap to allow uniform decomposition. Depending on the condition of the weather and the type of raw materials used, preparation of compost takes 4 - 6 months. High temperature and high humidity favour rapid decomposition. Addition of small quantities of urea and triple superphosphate hastens the rotting of raw materials like straw, sugarcane trash, rice husk etc. which decomposes very slowly. Microbial accelerator (e.g. *Trichoderma*), if available, may also be used for rapid composting; this is called trichocompost.

In composting, earthworms can play a good role. Earthworms convert organic wastes such as manure or household refuse to valuable compost. This is known as **vermicompost**. Earthworm inhabits organic matter lying on soil surface; eat fallen leaves and other non-decomposed litter. It has also been found to be especially efficient in breaking down the toughest organic wastes like sugarcane trash. Epigeic earthworms are most widely used in vermicomposting, owing to their tolerance to a wide range of temperature and moisture and consumption of large amounts of organic wastes daily. The species most often used for composting include *Eisenia fetida* (red wiggler or tiger worm), then others viz. *Eisenia andrei, Eudrilus eugeniae* and *Lumbricus rubellis.*

Bio-slurry

Bio-slurry is an anaerobically decomposition product of animal manure. Commonly used cowdung or poultry manure is an aerobically decomposition product. After extraction of biogas (chiefly CH_4), bio-slurry comes out of digester. Bio-slurry can be a potential source of organic matter. Fertilizer value of original manure (cowdung, poultry manure) is not hampered when it is turned to bio-slurry. Bio-slurry pit should be shaded to protect it from scorching sunshine and rains. A major limitation of the utilization of bio-slurry is to carry it from the pit to the distant crop field.

Green manure

Green manure (GM) refers to crops that are grown and ploughed down at the appropriate stage of growth. In some countries, farmers collect fresh leaves from the forests and apply to the soil. This is called green leaf manure. Green manure adds substantial quantities of organic matter and N to soils.

Any herbaceous plant may be used for green manuring, but plants of the family leguminosae are preferred because of the added advantage of getting fixed N. The common GM plants include dhaincha *(Sesbania aculeata),* African dhaincha *(S. rostrata),* sunhemp (*Crotalaria juncea*), cowpea, grasspea, soybean, mungbean, blackgram etc. The crops should be ploughed down when the plants are 50-55 days old. Rhizobial inoculation would be useful to obtain higher biomass in a given period over uninoculated legumes. Dhaincha needs to be incorporated to soil within a week before T. Aman rice planting. A green manure crop may add 10 - 15-ton biomass (fresh weight) per hectare and 60-120 kg of N/ha to the soil. *Azolla* (water fern) can also be used as a green manure in Boro rice field.

Carbon sequestration

Carbon sequestration (carbon sink) is a biochemical process by which atmospheric carbon is absorbed by living organisms including trees, soil microorganisms and crops and involving the storage of carbon in soils with the potential to reduce atmospheric CO_2 levels.

Carbon sequestration is capturing and securely storing carbon dioxide emitted from the global energy system. A large amount of carbon is stored in soils and vegetation, which is our natural carbon sink. Increasing carbon fixation through photosynthesis, slowing down or reducing decomposition of organic matter, and changing land use practices can enhance carbon uptake in these natural sinks. Terrestrial carbon sequestration is the process through which carbon dioxide (CO_2) from the atmosphere is absorbed by trees, plants and crops through photosynthesis, and stored as carbon in biomass (tree trunks, branches, foliage and roots) and soils. Therefore, a carbon sink occurs when carbon sequestration is greater than carbon releases over some time period.

Soils of Bangladesh have low reserves of organic carbon due to increasing cropping intensity, higher rate of organic matter decomposition under sub-tropical humid climate, low use of organic manure and little or no use of green manures. The highest depletion of soil carbon has been reported in soils of Meghna river floodplain (35%), followed by Madhupur Tract (29%), Brahmaputra floodplain (21%), Old Himalayan piedmontplain (18%) and Gangetic floodplain (15%). The sequestration of atmospheric C in the soil and biomass reduces greenhouse effect. Carbon sequestration is essential to improve soil quality, increase agronomic productivity and use efficiency of inputs like fertilizers and water and thus helps maintain or restore the capacity of soil to perform its production and environmental functions on a sustainable basis.

Carbon sequestration reflects the long-term balance between additions of organic C from different sources and its losses from soil. Following the adoption of large-scale intensive cropping, with the introduction of modern varieties and increased use of chemical fertilizers, the long-term balance would be modified. Intensive cropping encourages oxidative losses of C due to continued soil disturbance, while it also leads to a large-scale addition of C to the soil through crop residues. This may result in a buildup or depletion of soil carbon stock.

5.2 Integrated Nutrient Management

The basic concept of Integrated Nutrient Management (INM) is 'the management of all available plant nutrient sources, organic and inorganic, to provide optimum and sustainable crop production conditions within the prevailing farming system'. Therefore, the INM approach refers to an appropriate combination of mineral fertilizers, organic manures, crop residues, compost, N-fixing crops and bio-fertilizers taking into account of the local ecological conditions, land use systems and the individual farmer's social and economic conditions. In the INM practice it is important to consider the cropping pattern, not the single crop. Manure or fertilizer alone cannot sustain soil fertility and crop yield over time, their combination (i.e. INM) is essential for it.

The INM based fertilizer requirement can be calculated as follows:

$$A = B - C$$

A = Amount of nutrient needed from fertilizer source
B = Total need of a nutrient for a crop on soil test or AEZ basis
C = Amount of nutrient supply from manure application

Example: *Nutrient addition from manure and fertilizers for potato*

		N	P	K	S	Mg	Zn	B
Recommended dose (kg/ha)		135	30	90	10	-	2	1
Supply from manure	Cowdung t/ha	30	10	20	-	-	-	
Supply from fertilizer		105	20	70	10	-	2	1

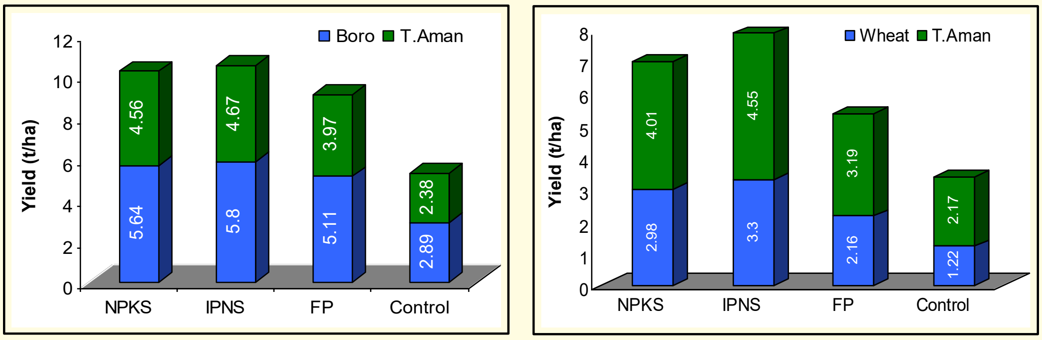

Fig. 5 Effects of different types of nutrient management on the yield of crops

5.3 Biochar

Biochar is a carbon-rich product generated from the pyrolysis of waste biomass under an oxygen-limited condition at relatively low temperature (<700°C). Waste biomass includes sawdust, rice husk, rice straw, wheat straw, sugarcane bagasse, animal litters etc. Biochar application has multiple benefits. It helps carbon sequestration, improvement of soil properties (increase of water holding capacity, decrease of soil acidity, increase of CEC, increase of N, P & K contents, reduction of CO_2 emission, and improvement of crop yield. The method for biochar production includes physical activation, chemical activation, and blending modification, which can improve the properties of biochar such as surface area, pore structure, functional groups, and nutrient contents.

Biochar has high surface area and porous structure, which allows it to retain moisture and nutrients in the soil, and its ability to act as a habitat for beneficial soil microbes. This can be especially useful in areas having light soil texture and very low soil organic matter. Another benefit is that it can help reduce greenhouse gas emissions. When organic matter is heated to create biochar, it undergoes a process called pyrolysis, which releases carbon dioxide. However, the carbon in the biochar is then sequestered in the soil, where it can remain for hundreds of years. So, biochar has the potential to offset some of the carbon emissions that released during the pyrolysis process.

When producing biochar for long-term carbon sequestration in soils, carbon content is almost the sole consideration, but when producing biochars to improve agricultural productivity, the carbon content of the biochar is not the main focus. Other factors of greater importance include structural characteristics and ion exchange capabilities. Due to its aromatic structure, biochar carbon is also chemically and biologically more stable than carbon in the original biomass. This has important implications for carbon sequestration.

Despite of its some potential advantages, biochar use has some disadvantages such as low sorption ability for heavy metals and organic contaminants. It is not commercially available in the market in Bangladesh as because it is still at research stage. Biochar production requires specialized equipment and facilities, which could be costly and unaffordable to smallholder farmers. Additionally, there is still a lack of research on the long-term effects of biochar on soil health indicating that its efficacy is not yet fully understood. Biochar treatment has a detrimental impact on the growth of heterotrophic soil microorganisms (depend on organic matter for both carbon and energy), many of them are beneficial (*Rhizobium, Azotobacter, Azospirillum* etc.). For this reason, the annual rate of biochar addition to soil should not exceed 2 tons per hectare. Biochar addition enhances CH_4 emissions by 10–14.0%.

5.4 Liming

Liming is done to correct soil acidity. Liming eliminates the toxic effect of Al, Fe and Mn, and increases the availability of P, Mo, Ca and Mg. Liming stimulates mineralization of organic N and fixation of atmospheric N. It improves soil physical conditions e.g. soil structure. Liming is generally practiced for dry land crops e.g. maize, wheat, grain legumes, oil seeds etc. This practice also reduces N_2O emissions, but this is more than offset by CO_2 emissions from the lime as it neutralizes acidity. Because crop plants vary in their tolerance to acidity and plant nutrients have different optimal pH ranges, target soil pH values could be set at 6.5 -7.0.

Limimg materials are calcium carbonate ($CaCO_3$), calcium oxide (CaO), calcium hydroxide [Ca(OH)2], magnesium carbonate ($MgCO_3$) and dolomite (dololime). Dololime ($CaCO_3$. $MgCO_3$) is now commonly used in this country. Wood ash, press mud from sugar mills, lime sludge from paper mills and sludge from water treatment plants can also be used as limimg materials.

Liming reactions begin with the neutralization of H^+ in the soil solution by either OH^- or HCO_3^- producing from liming materials. The continued removal of H^+ from the soil solution will ultimately result in the precipitation of free aluminium and iron into insoluble hydroxide forms and their replacement on soil exchange sites with basic cations. Consequently, the soil pH increases with increasing percent base saturation. In Bangladesh dolomitic lime, $CaMg(CO_3)_2$ (called dolochun, imported from Bhutan) is commonly used. Liming increases Ca^{2+} concentrations in the soil solution and thus increases soil pH through the reaction: $CaCO_3 + H^+ = Ca^{2+} + CO_2 + H_2O$. The amount of liming necessary to neutralize soil acidity depends on the soil pH, organic matter and clay contents. Liming is applicable to soils having pH below 5.5 at rate of 1-2 t/ha as dololime. After lime application it takes about 3 months' time to attain a maximum soil pH. Liming should be done in dry season at 3-4 weeks before the next crop is grown. Further liming may not be required within three years.

5.5 Conservation Agriculture

According to FAO (2006), Conservation Agriculture (CA) is a concept for resource-saving agricultural crop production that strives to achieve acceptable profits together with high and sustained production levels while concurrently conserving the environment. CA is based on enhancing natural biological processes above and below the ground. This farming system can prevent losses of soil fertility while regenerating degraded soil fertility. It is a promising approach in Bangladesh agriculture having environmental and economic benefits.

Conservation Agriculture (CA) stands on three pillars – Minimum tillage, Crop residue retention and Suitable crop rotations. Minimum tillage refers to minimum soil disturbances which could be zero tillage, strip planting or bed planting. For residue retention, crops will be cut at certain height, may be 30 cm for rice. Crop rotation could be preferably legume based. The 8th World Congress of Conservation Agriculture Declaration (2021) advocated for transformation of paddy rice systems into CA systems with legumes and cover crops.

There are numerous success stories of CA practice leading to sustained soil health, improved crop yields, reduced GHG emissions and cost savings. In Bangladesh, Conservation Agriculture (CA) is still a relatively new approach for intensively cultivated (3 crops/year) rice-based cropping systems that delivers significant crop produce and residues annually. Recently comprehensive field studies had been completed under the umbrella of CA based NUMAN (nutrient management) project funded jointly by ACIAR and KGF. One of these studies reveal that after 9 years, CA practice with high residue retention under strip tillage, the crops had higher N use efficiency, crop yield, system yield and increased annual gross margin in the rice-wheat-mungbean cropping system while the N fertilizer requirement increased minimally. By eliminating the need for tillage, farmers can save fuel costs and reduce carbon footprint. This can lead to more productive and sustainable farms as a long-term effect.

Despite many benefits, there are some limitations to CA. The one limitation is that the benefits of CA are not usually visible before five years which could be discouraging to many farmers. Another limitation is the increased reliance on herbicides, as the lack of tillage can lead to increase weed growth. Further potential limitation is the initial cost of implementing conservation agriculture; as specialized machinery is expensive. However, these costs may be offset by the long-term economic and environmental benefits of the system Nevertheless, the smallholder farmers can't afford the cost of CA machineries. There are also challenges to uptake and scaling up of the CA practice. The government strategies towards pertinent demonstration and training along with soft loan could help for farmers' realization of CA benefits and thereby wide scale adoption of this noble approach may be possible in the country.

5.6 Organic Agriculture

Conventional Agriculture focuses merely on yield benefits with an intensive use of agro-chemical inputs. The global concern over the excessive use of agro-chemicals influences the organic methods of agricultural production. Organic farming is an emerging practice of crop production that concerns not to use pesticides, fertilizers, genetically modified organisms and growth hormones. Among the pesticides, insecticides account for about 80%, and are largely used for vegetables farming. Organic farming system relies on crop rotations, crop residues, animal manures, legumes, green manures, safe off-farm organic wastes and aspects of biological pest control. Advantages of organic farming include recycling of nutrients, sequestration of carbon, reducing chemical pollution of soil & water, promotion of biological diversity and providing safe & healthy food. Nevertheless, it has some disadvantages which could be yield loss, higher price of produce and limited size of market.

Organic Agriculture (OA) is quite applicable to homestead gardening which covers a small area. Cereal and forage crops can be grown organically easily due to relatively less pest infestation and also low nutrient requirement. Fruit and vegetable crops present greater challenges. Some pest and disease problems are difficult to manage by organic methods. It is well agreed that neither fertilizer nor manure alone can sustain soil fertility and crop productivity, their integration is essential to achieve sustainable crop yield without incurring loss to soil fertility. Organic farming is a holistic production management system which promotes healthy agro-ecosystem, including biodiversity, biological cycles, and soil biological activity, as opposed to using synthetic materials. To promote organic farming wider participation of policy makers, growers, extension personnel and business people is imperative.